# The t Test for a Population Mean

Lectures 22–23

QBA201 – Quantitative Business Analysis

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# What changes when we estimate the spread?

## Recall the denominator

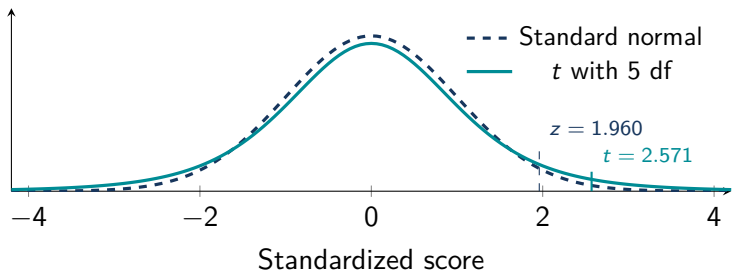
For a mean test, the numerator is the observed mean minus the benchmark: $\bar{x} - \mu_0$.

| Population SD | Standard error used | Reference under $H_0$ |
|---|---|---|
| Known $\sigma$ | $\sigma/\sqrt{n}$ | Standard normal $Z$ |
| Unknown $\sigma$ | $s/\sqrt{n}$ | Student $t$ with $n-1$ degrees of freedom |

For independent normal observations, $\bar{X}$ stays normal. The statistic with the estimated standard error follows $t$.

# A small sample needs a wider reference

If $n = 6$, the $t$ reference has five degrees of freedom. Compare the upper cutoffs leaving 2.5% on their own curves.




The $t$ tails are heavier. As degrees of freedom increase, the curve and its cutoffs approach the standard normal.

# Build a t score for an average cost

An independent random sample of 25 normal parcel-handling costs gives mean AED 18.80 and sample SD AED 2.00. Test whether the population mean differs from AED 18. Use $\alpha = .05$.

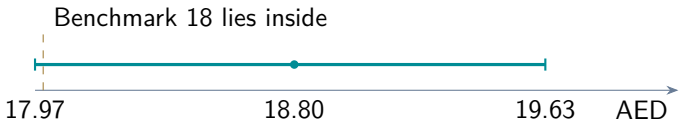
$$H_0 : \mu = 18, \qquad H_1 : \mu \neq 18.$$

## Calculate the gap in estimated standard errors

$$\widehat{\mathrm{SE}} = \frac{2}{\sqrt{25}} = 0.40, \qquad t_{\mathrm{obs}} = \frac{18.80 - 18}{0.40} = 2.00, \qquad \nu = 24.$$

# Check the decision against the interval

For the two-sided question, $p = 2\,\mathrm{P}(T_{24} \geq 2) = 0.0569$. At 5%, fail to reject $H_0$.




95% interval : $\quad 18.80 \pm 2.064(0.40) \approx [17.97, 19.63]$.

The interval includes AED 18. The data leave the difference unresolved at 5%.

# Use the signed t score for a directional claim

With 24 degrees of freedom, consider three separate preplanned questions for $t_{\text{obs}} = 2.00$.

| Alternative | Area under $t_{24}$ | p-value | At 5% |
|---|---|---|---|
| $\mu > \mu_0$ | Right of 2.00 | .0285 | Reject |
| $\mu < \mu_0$ | Left of 2.00 | .9715 | Fail to reject |
| $\mu \neq \mu_0$ | Both beyond $\pm 2.00$ | .0569 | Fail to reject |

A negative score of the same size reverses the one-tail areas. Use the alternative chosen before sampling.

# Check whether the sampling design changes the SE

## Independent normal observations

For an unknown population SD, $s/\sqrt{n}$ is the denominator for the exact one-sample $t$ result.

## Sample without replacement from a fixed list

For a simple random sample of $n$ from a list of $N$ units, a design-based estimated SE is

$$\frac{s}{\sqrt{n}}\sqrt{1-\frac{n}{N}}.$$

The adjustment reduces estimated variability. The independent-normal $t$ reference needs its own assumptions.

# Try a left-tail t test for delivery time

A logistics firm wants to know whether its mean delivery time is below 45 minutes. For an independent random sample of 16 delivery times from a normal population, $\bar{x} = 42.5$ and $s = 5.0$ minutes. Use $\alpha = .05$.

## Choose the tail and the reference

$$H_0 : \mu \geq 45, \qquad H_1 : \mu < 45, \qquad t_{\text{obs}} = \frac{42.5 - 45}{5/\sqrt{16}} = -2.00, \qquad \nu = 15.$$

## Check the business decision

The left-tail area is $\mathrm{P}(T_{15} \leq -2.00) = .0320$. Reject at 5%. The sample supports a population mean delivery time below 45 minutes.